

Question:

An 83-year-old male presents to the emergency department with altered mental status. The patient’s vitals signs are as follows: temperature is 100.7 deg F (38.2 deg C), blood pressure is 143/68 mmHg, heart rate is 102/min, and respirations are 22/min. The caretaker states that the patient is usually incontinent of urine, but she has not seen any soiled adult diapers in the past 48 hours. A foley catheter is placed with immediate return of a large volume of cloudy, pink urine. Which of the following correctly explains the expected findings from this patient’s dipstick urinalysis? Answer Choices: A. Detection of an enzyme produced by red blood cells B. Direct detection of white blood cell surface proteins C. Detection of urinary nitrate conversion by gram-negative pathogens D. Detection of an enzyme produced by gram-negative pathogens

Reasoning + Answer:

<think> The question is asking about the expected findings from the patient's dipstick urinalysis. To answer this question, we need to identify the key information provided in the scenario and use it to determine the most likely explanation for the expected findings. The patient has a Foley catheter inserted with immediate return of a large volume of cloudy, pink urine. This suggests that the patient has a urinary tract infection (UTI). The dipstick urinalysis is used to detect various compounds in the urine, including leukocyte esterase, which is an enzyme produced by white blood cells, and nitrate, which is converted by gram-negative pathogens. Given the presence of cloudy, pink urine, it is likely that the patient has a UTI caused by gram-negative pathogens, which would result in the detection of urinary nitrate conversion by these pathogens. </think> <answer> Detection of urinary nitrate conversion by gram-negative pathogens </answer>

